Dystonia 25 (DYT25 / DYT-GNAL) — Comprehensive Disease Characterization

MONDO: MONDO:0014033 · **OMIM:** 615073 · **Gene:** *GNAL* (Gα-olf) · **Category:** Mendelian (autosomal dominant) **Report date:** 2026-10-20 · Evidence base: human clinical pedigrees, rodent models, in vitro functional assays.

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1. Disease Information

Overview. Dystonia 25 (DYT25; now designated **DYT-GNAL**) is an autosomal dominant, adult-onset **isolated (primary) dystonia** caused by loss-of-function mutations in *GNAL*. Dystonia is a movement disorder defined by sustained or intermittent involuntary muscle contractions producing twisting, repetitive movements and abnormal postures. DYT25 characteristically presents as **focal or segmental cranio-cervical dystonia** — most often cervical dystonia (torticollis) — that may spread to oromandibular, laryngeal (spasmodic dysphonia), and upper-limb regions, frequently with dystonic tremor. *GNAL* was the **first gene definitively confirmed to cause common adult-onset focal/segmental isolated dystonia** (Fuchs et al. 2013,

P 23222958

; Vemula et al. 2013,

P 23449625

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Key identifiers. - OMIM: **615073** (Dystonia 25, DYT25) - MONDO: **MONDO:0014033** - Orphanet: dystonia 25 falls under ORPHA:98806 (isolated dystonia) / adult-onset isolated focal dystonia spectrum - Gene (HGNC): **GNAL**, HGNC:4388; NCBI Gene ID 2774; UniProt P38405 (GNAL_HUMAN) - MeSH: Dystonic Disorders (D020821); Dystonia (D004421) - ICD-10: **G24** (Dystonia; G24.3 spasmodic torticollis, G24.4 idiopathic orofacial dystonia); ICD-11: **8A02** (Dystonia)

Synonyms / alternative names: DYT25; DYT-GNAL; dystonia-25; primary torsion dystonia 25; GNAL-related dystonia; adult-onset cervical/cranio-cervical dystonia (GNAL type).

Information source type. Disease-level knowledge is derived from **aggregated resources** (OMIM, Orphanet) and from **individual-patient case series / pedigrees** reported in primary literature; mechanistic data come from rodent models and in vitro assays rather than EHR aggregation.

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2. Etiology

Primary cause — genetic. Heterozygous **loss-of-function mutations in *GNAL***, encoding the stimulatory G-protein alpha subunit **Gα-olf** (G(olf)). The disorder is monogenic and autosomal dominant with **incomplete penetrance** (Vemula 2013, P 23449625: "These mutations show incomplete penetrance").

Genetic risk factors (causal variants). Nonsense, frameshift, splice-site, start-loss, and missense variants across the *GNAL* coding sequence (see §4). There are no established common susceptibility (GWAS) loci; DYT25 is a Mendelian, not polygenic, entity.

Environmental risk factors. None established as causal. As with other focal dystonias, non-specific triggers/modifiers hypothesized for adult-onset focal dystonia generally (e.g., repetitive/occupational activity, peripheral trauma, prior neck injury for cervical dystonia) may act as triggers in genetically predisposed individuals, but no GNAL-specific environmental factor is proven. **Age** (adult onset) and possibly **sex** modulate expression.

Protective factors. None genetically defined. Incomplete penetrance implies the existence of unidentified genetic/environmental **modifiers** that can prevent clinical expression in carriers, but these are not characterized.

Gene–environment interactions. The incomplete, age-dependent penetrance of *GNAL* variants is consistent with gene–environment and gene–gene interactions shaping whether and when a carrier manifests dystonia; specifics remain unknown.

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3. Phenotypes

DYT25 is an **isolated** dystonia — dystonia is the only motor feature (± tremor), without other neurological signs, and brain MRI is typically normal.

Phenotype	Type	HPO term	Onset / severity / progression	Frequency
Cervical dystonia (torticollis)	Clinical sign	**HP:0002317** (Torticollis) / HP:0030972 (Cervical dystonia)	Adult-onset (often 3rd–6th decade); moderate; usually focal, limited spread	Most common site; typical presenting feature
Cranial / oromandibular dystonia	Clinical sign	**HP:0002540**? — use HP:0012048 (Oromandibular dystonia)	Adult-onset; variable	Common in segmental cases
Laryngeal dystonia (spasmodic dysphonia)	Clinical sign	HP:0001618 (Dysphonia)	Adult-onset	Reported (Vemula 2013; Carecchio 2016)
Blepharospasm / facial dystonia	Clinical sign	HP:0000643 (Blepharospasm)	Adult-onset	Occasional (cranial involvement)
Upper-limb / hand-forearm (writer's) dystonia	Clinical sign	HP:0002527? use HP:0003470 / HP:0100660 (Dystonic limb)	Adult-onset	Reported
Dystonic tremor (head/arm)	Clinical sign	HP:0002322 (Resting tremor)/ HP:0001337 (Tremor)	Variable; may mimic DYT11/ essential tremor	Prominent in some families (Carecchio 2016)
Generalization of dystonia	Clinical course	HP:0007325 (Generalized dystonia)	Rare; generally does NOT generalize	Uncommon (one Chinese case generalized)
Microsmia / hyposmia	Symptom/ lab	HP:0004409 (Hyposmia)	—	Reported in p.V228F carriers (Gα-olf role in

Phenotype	Type	HPO term	Onset / severity / progression	Frequency
				olfaction), not consistent
Psychiatric / "mental" symptoms	Behavioral	HP:0000708 (Behavioral abnormality)	—	Rare (Fan 2022, PMID 35396637)

General phenotype characteristics. *Onset:* adult-onset (typically ~ 4th–5th decade; range childhood to late adult). *Severity:* mild-to-moderate, variable even **within families (intra-familial heterogeneity)** (Carecchio 2016, PMID 26725140). *Progression:* slowly progressive then **plateaus**; focal→segmental spread possible but **generalization is rare**; course is chronic/lifelong, non-remitting (dystonic tremor can fluctuate). *Distribution:* often asymmetric.

Quality-of-life impact. Cervical and cranial dystonia cause pain, social embarrassment, impaired head control, speech/swallowing difficulty, and functional disability; QoL is reduced similarly to other adult-onset focal dystonias (disease-specific tools such as CDIP-58 / TWSTRS for cervical dystonia; no GNAL-specific QoL study). Botulinum toxin and DBS substantially improve QoL (§12).

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4. Genetic / Molecular Information

Causal gene. ***GNAL*** (guanine nucleotide-binding protein, alpha-activating activity polypeptide, olfactory type) — chromosome **18p11.21**; HGNC:4388; OMIM gene *139312*; encodes **Gα-olf (G(olf) subunit alpha)**, UniProt P38405.

Representative pathogenic variants (all heterozygous, germline). - **p.Ser293*** (nonsense) — original family (Fuchs 2013, PMID 23222958) - **p.Val137Met** (missense) — original family (Fuchs 2013) - **p.Val228Phe** (c.682G>T, missense) — African-American pedigree; carriers showed microsmia (Vemula 2013,

)

P 23449625

- **p.Arg198Thrfs*13** (c.591dupA, frameshift); **p.Arg245*** (c.733C>T, nonsense); **p.Met1?** (c.3G>A, start-loss) (Vemula 2013) - **p.Asp210Asn** (c.628G>A, exon 9, missense) — Italian family (Carecchio 2016,

)

P 26725140

- **p.Val354Ala** (c.1061T>C, missense) — *de novo* in sporadic Serbian cervical dystonia patient (Dobričić 2014,

)

P 24729450

- **p.Ser95*** (c.284C>T, nonsense) and **c.932-7T>G** (splice) — Chinese patients (Miao 2013,

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P 23759320

Variant classification (ACMG/AMP). Truncating variants (nonsense, frameshift, canonical splice, start-loss) are classified **pathogenic** (loss of function is the established mechanism for a haploinsufficient gene). Missense variants are often **likely pathogenic/VUS**, supported by functional assays (BRET showing impaired Gα-olf coupling; Fuchs 2013), in silico prediction, and segregation.

Variant types/classes. Missense, nonsense, frameshift, splice-site, and start-loss — i.e., a mix consistent with **haploinsufficiency**. In a representative set of 10 reported variants, **~60% are truncating** (nonsense/frameshift/start-loss/splice) and **~40% missense**, supporting loss-of-function rather than gain-of-function or dominant-negative mechanisms (see `gnal_variant_spectrum.png`).

Classification note. GNAL causes **isolated dystonia** (dystonia as the sole motor feature apart from tremor; MDS 2013 axis-I classification), distinct from combined dystonia-parkinsonism (e.g., ATP1A3/DYT12) or myoclonus-dystonia (SGCE/DYT11). However, GNAL "and CIZ1 ... are probably not a common cause of classic cervical dystonia" (Balint & Bhatia 2015,

)

P 25643588

— important for setting genetic-testing yield expectations.

Allele frequency. Pathogenic *GNAL* variants are **rare/absent in gnomAD** controls. In dystonia cohorts, mutation frequency is low: **~0.4%** in Serbian cranio-cervical dystonia (Dobričić 2014) and **~3.4%** among Chinese cervical-onset/involved primary dystonia after excluding TOR1A/THAP1 (Miao 2013). GNAL accounts for a small but meaningful fraction of adult-onset isolated dystonia.

Somatic vs germline. Germline (including documented *de novo* germline events — Dobričić 2014).

Functional consequence. Loss of function / haploinsufficiency. BRET assays confirmed impaired function of several mutants (Fuchs 2013: "Impaired function of several of the mutants was shown by bioluminescence resonance energy transfer (BRET) assays"). Reduced Gα-olf lowers D1-/A2A-receptor–stimulated adenylyl cyclase activity in striatum.

Modifier genes. Unidentified; incomplete penetrance implies modifiers exist.

Epigenetic information. Gα-olf participates in **DALN/histone H3 phosphorylation** downstream of dopamine signaling in striatum (noted by Vemula 2013), linking it to activity-dependent transcription; no disease-specific methylation signature reported.

Chromosomal abnormalities. None characteristic; DYT25 is caused by point mutations/small indels, not large structural variants.

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5. Environmental Information

- **Environmental factors:** None established as causal for DYT25. No toxin/radiation/occupational exposure is proven to cause GNAL dystonia.
- **Lifestyle factors:** None established. (For focal dystonias broadly, repetitive task performance and peripheral trauma are debated triggers.)
- **Infectious agents:** Not applicable — DYT25 is a genetic, non-infectious disorder.

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6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. A heterozygous **loss-of-function *GNAL* mutation** (nonsense/frameshift/splice/missense) **results in** reduced quantity or function of **Gα-olf** protein (haploinsufficiency; demonstrated by BRET, Fuchs 2013).
2. Because Gα-olf is the principal **Gαs-like subunit coupling D1 dopamine and A2A adenosine receptors to adenylyl cyclase in the striatum**, reduced Gα-olf **leads to**

decreased receptor-stimulated **cAMP production** in striatal spiny projection neurons (SPNs) (Vemula 2013; Yu-Taeger 2020).

3. Diminished cAMP/PKA signaling **results in** blunted **dopamine- and adenosine-mediated signaling and synaptic plasticity** in both direct (D1) and indirect (A2A) pathway SPNs (demonstrated in a GNAL rat model, Yu-Taeger 2020,

). P 31678405

4. In parallel, **striatal cholinergic interneurons become dysfunctional** (abnormal excitability/acetylcholine release) — a convergent node shared across genetic dystonias (Eskow Jaunarajs 2019,

). P 31320448

Inferred to amplify the plasticity deficit.

5. Altered dopaminergic/cholinergic balance **leads to loss of corticostriatal long-term depression (LTD)** — a signature plasticity defect also seen in DYT1 (Martella 2021,

). P 33991565

This is **demonstrated** and is **pharmacologically reversible** (chronic mGlu5 negative allosteric modulator dipraglurant restored LTD).

6. Disordered basal-ganglia output **propagates to and dysregulates the cerebello-thalamo-cortical network**, producing **abnormal/pathological firing patterns** (demonstrated in DYT25 mice: functional abnormalities in cerebello-thalamic pathways, Aïssa 2022,

). P 35699413

Branch: both basal-ganglia and cerebellar circuits contribute (network model of dystonia).

7. Aberrant motor-network firing **results in** excessive co-contraction of agonist/antagonist muscles → **sustained involuntary contractions, abnormal postures, and dystonic tremor** — the clinical phenotype.

Biochemical grounding of Gα-olf

Gα-olf (G(olf) subunit alpha) is a **stimulatory, Gs-like G-protein alpha subunit** that couples GPCRs to **adenylyl cyclase III** to raise cAMP. Classic biochemistry (Jones et al. 1990,

) P 2105931

showed Golf is "functionally and structurally similar" to Gs, interacts with β-adrenergic receptors (activated by isoproterenol, EC50 ≈ 120 nM) and is constitutively activated by

cholera toxin. Its signaling is amplified by the GTP-exchange factor **Ric-8B**, which potentiates Gα-olf-dependent cAMP accumulation and is co-expressed with Gα-olf in olfactory neurons and brain (Von Dannecker et al. 2005, P 15829631) — a candidate modifier/co-factor of residual Gα-olf activity in DYT25. In the striatum, Gα-olf substitutes for Gαs as the principal stimulatory subunit, explaining why its haploinsufficiency selectively blunts striatal D1/A2A-cAMP signaling.

Category detail

- **Molecular pathways: GPCR → Gα-olf → adenylyl cyclase (AC3/AC5) → cAMP → PKA → DARPP-32** signaling; dopamine (D1/D5) and adenosine (A2A) receptor cascades. KEGG: dopaminergic synapse (hsa04728), cAMP signaling (hsa04024), GABAergic/cholinergic synapse. Reactome: G-alpha (s) signalling events.
- **Cellular processes:** Synaptic plasticity (LTD/LTP), neuronal excitability, neurotransmission; cholinergic interneuron hyperexcitability; activity-dependent transcription (histone H3 phosphorylation).
- **Protein dysfunction:** Reduced Gα-olf abundance/coupling (loss of function); truncating alleles likely degraded (NMD); missense alleles impair nucleotide exchange/effector coupling.
- **Metabolic changes:** Lowered cyclic-AMP second-messenger tone in striatum (not a systemic metabolic disorder).
- **Immune involvement:** None established.
- **Tissue damage mechanisms: None/neurodegeneration absent** — dystonia is a *functional/circuit* disorder ("lack of overt neuropathology," Weisheit 2018, P 29325615). No cell death, gliosis, or structural lesion.
- **Biochemical abnormality:** Deficient D1/A2A-receptor-coupled adenylyl cyclase activation; impaired cAMP signaling.
- **GO terms:** GO:0007186 (GPCR signaling), GO:0007189 (adenylate cyclase-activating GPCR signaling / cAMP), GO:0007212 (dopamine receptor signaling), GO:0060079/GO:0048167 (regulation of synaptic plasticity), GO:0007608 (sensory perception of smell).

- **Cell types (CL):** CL:0000029/striatal medium spiny neuron (direct/indirect pathway SPN), CL:0000108 (cholinergic neuron / striatal cholinergic interneuron), cerebellar neurons, thalamocortical neurons.

Human imaging corroboration. Resting-state fMRI in the cervical-dystonia phenotype independently confirms **functional (network) dysfunction without structural damage** across the cerebello-basal ganglia-thalamo-cortical loop (Giannì 2022, ; P 35169015 Zito 2022, ; P 34875561 Norris 2020,), P 32913023 and GPi-DBS normalizes this network connectivity toward healthy-control levels (Filip 2025,) P 39986202 — consistent with the rodent-derived mechanism. (These cohorts are predominantly idiopathic, not genotyped GNAL; link is by phenotype.)

Upstream vs downstream: GNAL LOF and striatal cAMP deficit are **upstream**; corticostriatal LTD loss and cerebello-thalamo-cortical dysfiring are **downstream**; muscle co-contraction is the terminal output.

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7. Anatomical Structures Affected

- **Organ level: Brain / central nervous system** (nervous system) — functionally affected; no peripheral organ pathology.
- **Primary structures: Striatum** (caudate-putamen) — the main site of Gα-olf expression and dysfunction. UBERON:0002435 (striatum); UBERON:0001874 (putamen); UBERON:0001873 (caudate).
- **Secondary / network structures: Globus pallidus internus** (DBS target; UBERON:0002477), **thalamus** (UBERON:0001897), **cerebellum** (UBERON:0002037), **motor cortex** (UBERON:0001384) — the cerebello-thalamo-cortical and cortico-striato-pallidal motor loops.

- **Clinically affected effectors (dystonic muscles):** cervical (sternocleidomastoid, splenius, trapezius), cranial/oromandibular, laryngeal, upper-limb muscles.
- **Tissue/cell level:** Nervous tissue; striatal **medium spiny neurons** and **cholinergic interneurons** (CL terms in §6).
- **Subcellular level (GO Cellular Component): Plasma membrane** (GO:0005886) and heterotrimeric G-protein complex (GO:0005834) where Gα-olf couples GPCRs to adenylyl cyclase; cytoplasmic/membrane signaling compartments. No organelle-specific storage pathology.
- **Localization / lateralization:** Bilateral basal ganglia involvement at the network level; **clinical manifestation is often asymmetric** (e.g., laterocollis/torticollis to one side; asymmetric limb tremor).

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8. Temporal Development

- **Onset: Adult-onset** — typically **4th–6th decade** (range from childhood to late adulthood). Onset is **insidious/chronic**, not acute.
- **Progression: Slowly progressive** over months–years, usually reaching a plateau; focal→segmental **spread** can occur (cervical → cranial/laryngeal/limb) but **generalization is rare**. Course is **chronic and lifelong**, non-remitting (dystonic tremor may fluctuate).
- **Stages:** No formal staging. Clinically: focal → segmental (± tremor), stabilizing.
- **Remission:** Spontaneous remission is uncommon; symptomatic control via botulinum toxin / DBS (treatment-induced improvement, not cure).
- **Critical periods:** Age-dependent penetrance means the **adult window** is when manifestation occurs; earlier intervention (shorter disease duration) predicts better DBS response (Tisch & Kumar 2020).

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9. Inheritance and Population

- **Inheritance: Autosomal dominant** (AD). HGNC *GNAL*, 18p11.21.
- **Penetrance: Incomplete, age-dependent** (Vemula 2013). Unaffected obligate carriers occur within pedigrees.

- **Expressivity: Highly variable**, including intra-familial heterogeneity in site and severity (Carecchio 2016).
- **De novo / mosaicism:** Documented **de novo** mutations cause seemingly sporadic cases (Dobričić 2014); germline mosaicism not specifically reported.
- **Anticipation:** Not applicable (not a repeat-expansion disorder).
- **Founder effects / consanguinity:** None established (dominant; multiple independent mutations across populations). Consanguinity not relevant.
- **Carrier frequency:** Pathogenic alleles are rare; no established population carrier frequency.
- **Epidemiology:** No DYT25-specific prevalence figure. Adult-onset focal dystonia overall affects ~15–30 per 100,000 (cervical dystonia the most common focal form). *GNAL* explains a **minority** of isolated dystonia: ~**0.4%** (Serbian cranio-cervical cohort, Dobričić 2014) to ~**3.4%** (Chinese cervical cohort after excluding TOR1A/THAP1, Miao 2013); roughly ~1–2% of familial PTD in mixed-European cohorts (Fuchs/Vemula 2013).
- **Population demographics / geographic distribution:** Reported across diverse ancestries — European/Caucasian, African-American, Serbian, Italian, Chinese, Korean — i.e., **no restricted geographic distribution**; mutations are private/family-specific.
- **Sex ratio:** No strong GNAL-specific skew reported (cervical dystonia generally has mild female predominance).
- **Age distribution:** Affected individuals cluster in adulthood (middle age).

10. Diagnostics

- **Clinical diagnosis** is primary: adult-onset isolated cranio-cervical/segmental dystonia ± tremor, normal cognition, and typically **normal brain MRI** (isolated dystonia, no structural lesion).
- **Laboratory tests / biomarkers:** No specific blood/urine biomarker; routine metabolic/copper (to exclude Wilson disease) and other labs are used for differential diagnosis, not for positive diagnosis. Olfactory testing may reveal subtle hyposmia in some carriers (not diagnostic).
- **Imaging:** Brain MRI is **normal/non-specific** (used to exclude secondary causes). Research **resting-state fMRI** of the cervical-dystonia phenotype shows **network, not structural**, abnormalities — loss of dentate–sensorimotor-cortex connectivity anticorrelation and

increased pallidal connectivity with preserved global architecture and normal volumes (Giannì 2022,

), P 35169015

somatotopic motor-cortex–cerebellum hyperconnectivity (Zito 2022,

), P 34875561

and regional striatal/sensorimotor connectivity reduction (Norris 2020,

). P 32913023

These are research tools, not clinical diagnostics, and cohorts are mostly idiopathic (not genotyped GNAL).

- **Electrophysiology:** EMG used to guide botulinum toxin injections and characterize dystonic vs tremor activity; not diagnostic of GNAL.
- **Genetic testing (confirmatory, gold standard):**
- **Approach:** After exclusion of acquired causes and often *TOR1A*/*THAP1*, test *GNAL* via **multigene dystonia NGS panels** or **whole-exome/whole-genome sequencing**; single-gene *GNAL* Sanger sequencing (coding + splice sites) for targeted confirmation and family segregation.
- WES/WGS: high yield in familial adult-onset dystonia; was the discovery method (Fuchs 2013; Vemula 2013).
- Gene panels: dystonia panels include *GNAL*, *TOR1A*, *THAP1*, *ANO3*, *KMT2B*, *GCH1*, etc.
- CMA/karyotype/FISH/mtDNA/repeat-expansion testing: **not indicated** (point-mutation disorder, not structural or repeat-based).
- **Clinical criteria / classification:** Per the 2013 MDS consensus classification of dystonia (axis I clinical, axis II etiology); GNAL = inherited isolated dystonia (**DYT-GNAL**).
- **Differential diagnosis:** Other isolated dystonias (**DYT-TOR1A/DYT1**, **DYT-THAP1/DYT6**, **DYT-ANO3/DYT24**, **DYT-KMT2B/DYT28**), dopa-responsive dystonia (**GCH1**), essential tremor (dystonic tremor can mimic; Carecchio 2016 noted partial mimicry of DYT11 myoclonus-dystonia), Wilson disease, tardive/drug-induced dystonia, functional dystonia. Distinguishing features: adult cranio-cervical onset, AD family history, normal MRI, and a *GNAL* variant.
- **Screening:** No population newborn screening. **Cascade genetic testing** of at-risk adult relatives and **genetic counseling** are appropriate given AD inheritance and incomplete penetrance.

11. Outcome / Prognosis

- **Survival / mortality:** DYT25 is **not life-shortening**; normal life expectancy. No disease-specific mortality.
- **Morbidity / disability:** Chronic motor disability, pain (especially cervical), speech/voice impairment (laryngeal involvement), and psychosocial impact; generally **non-progressive to generalization** and does not cause dementia or systemic decline.
- **Disease course:** Slowly progressive then stable; chronic lifelong.
- **Quality of life:** Reduced by motor symptoms and pain but **substantially improved by treatment** (botulinum toxin, DBS).
- **Recovery potential:** No spontaneous cure; excellent symptomatic control achievable — GPi-DBS gave ~47% early and ~63% late motor-score improvement in GNAL patients (Ahn 2019,

  P 31158945

).
- **Prognostic factors:** Shorter disease duration and isolated (vs combined) phenotype predict better DBS outcome (Tisch & Kumar 2020,

  P 33488508

).
  GNAL genotype itself is a **favorable predictor** of DBS response.

12. Treatment

No disease-modifying/curative therapy exists; management is symptomatic.

Pharmacotherapy (oral). - **Anticholinergics** (e.g., trihexyphenidyl) — rational given striatal cholinergic dysfunction; modest benefit (NCIT: Trihexyphenidyl). - Other oral agents used in dystonia: benzodiazepines (clonazepam — helpful for tremor/myoclonic features), baclofen, tetrabenazine (for select cases). Variable efficacy.

Chemodenervation (first-line for focal dystonia). - **Botulinum neurotoxin (BoNT-A/B)** injections into dystonic muscles (cervical, laryngeal, cranial) — mainstay for focal cervical/cranial/laryngeal dystonia; improves posture, pain, and voice (NCIT: Botulinum Toxin, C837). **Class I evidence, Level A recommendation** for cervical dystonia — increased range of neck

movement, decreased pain, increased functional capacity (Novak et al. 2010 international consensus,

20633181

).

Applies to DYT25 given its idiopathic-type cervical/cranial presentation.

Surgical / interventional (for medically refractory disease). - **Globus pallidus internus deep brain stimulation (GPi-DBS)** — highly effective in GNAL dystonia: **100% early responder rate (5/5)**, motor-score reduction **46.9% early / 63.4% late**, comparable to DYT1 and superior to DYT6 (Ahn 2019,

31158945

); excellent outcomes also in Carecchio 2016 (two cases) and reviewed by Tisch & Kumar 2020. NCIT: Deep Brain Stimulation.

Pharmacogenomics / experimental (preclinical). - **mGlu5 negative allosteric modulator dipraglurant** rescued corticostriatal LTD in GNAL rat and DYT1 mouse models (Martella 2021,

33991565

) — a mechanistically rational candidate, **not yet an approved DYT25 therapy**. - Targeting striatal cAMP/adenosine (A2A) signaling is a theoretical avenue from the mechanism.

Supportive / rehabilitative: Physiotherapy, occupational therapy, speech therapy (for laryngeal involvement), pain management, sensory-trick training, patient education.

Treatment algorithm: (1) confirm diagnosis + genetic counseling → (2) focal: botulinum toxin ± oral anticholinergics → (3) segmental/refractory: optimize oral meds → (4) refractory/disabling: **GPi-DBS** → (5) ongoing rehabilitation and symptom support.

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13. Prevention

- **Primary prevention:** None possible (monogenic, largely non-modifiable). No vaccine/behavioral intervention.
- **Secondary prevention / early detection: Genetic counseling and cascade testing** of adult at-risk relatives; awareness enables early botulinum toxin / earlier DBS consideration (shorter duration → better outcome).

- **Reproductive options:** Prenatal diagnosis and **preimplantation genetic testing (PGT-M)** are theoretically available for known familial mutations, but **incomplete penetrance and adult onset** strongly complicate counseling and uptake.
- **Immunization / public-health / environmental interventions:** Not applicable.
- **Tertiary prevention:** Prevent complications/disability through optimized symptom control, pain management, and rehabilitation.

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14. Other Species / Natural Disease

- **Taxonomy / orthologs:** *GNAL* is conserved; ortholog in mouse **Gnal** (NCBI Gene 14680; *Mus musculus*, NCBI:txid10090) and rat **Gnal** (*Rattus norvegicus*, NCBI:txid10116).
- **Natural disease in other species:** No well-documented naturally occurring *GNAL* dystonia in companion animals or wildlife (not in OMIA as an established natural model); DYT25 is studied via engineered/induced laboratory models rather than natural animal disease.
- **Comparative biology:** The striatal Gα-olf→adenylyl cyclase→cAMP signaling axis is evolutionarily conserved across mammals, underpinning translational validity of rodent models.
- **Zoonotic potential:** Not applicable (genetic disorder).

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15. Model Organisms

- **Mouse — *Gnal*$^{+/-}$ heterozygous knockout:** A **novel dystonia model** recapitulating haploinsufficiency; useful for pathophysiology (Pelosi et al. 2017,

P 28546310

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Cerebello-thalamic pathway dysfunction demonstrated in DYT25 mice (Aïssa 2022,

P 35699413

).
Cell-type-specific Gnal loss-of-function mice refine which neuronal populations drive the phenotype (Longueville et al. 2025,

P 40902679

).

- **Rat — GNAL model:** Shows **impaired dopamine-/adenosine-mediated signaling and plasticity** and **loss of corticostriatal LTD**, rescued by chronic mGlu5 NAM (Yu-Taeger 2020, P 31678405; Martella 2021, P 33991565).
- **Model types available:** Constitutive knockout/heterozygous, and **conditional/cell-type-specific** (Cre-driven) Gnal alleles.
- **In vitro / cellular:** Patient **lymphoblastoid cell lines** (p.V228F showed upregulated cell-cycle/development genes; Vemula 2013); heterologous **BRET assays** quantifying mutant Gα-olf coupling (Fuchs 2013).
- **Phenotype recapitulation:** Rodent models reproduce **electrophysiological/plasticity endophenotypes** (loss of LTD, impaired D1/A2A signaling, cerebello-thalamic abnormalities) more robustly than overt motor dystonia — a common limitation of dystonia models (overt sustained dystonic postures are often subtle).
- **Limitations:** Incomplete recapitulation of the human adult-onset focal motor phenotype; interspecies differences in penetrance and circuit organization.
- **Resources:** MGI (Gnal), RGD (Gnal), IMPC/IMSR for knockout lines.

Summary of Evidence Sources

- **Human clinical / genetic:** PMIDs 23222958 (Fuchs), 23449625 (Vemula), 24729450 (Dobričić), 26725140 (Carecchio), 23759320 (Miao), 31158945 (Ahn, DBS), 33488508 (Tisch & Kumar, DBS review), 35396637 (Fan), 29325615 (Weisheit review), 25643588 (Balint & Bhatia, classification/review).
- **Human neuroimaging (cervical-dystonia phenotype):** PMIDs 35169015 (Giannì), 34875561 (Zito), 32913023 (Norris), 39986202 (Filip, DBS network restoration), 40820429 (Xue, DBS connectomes).
- **Model organism / in vitro:** PMIDs 28546310 (Pelosi mouse), 31678405 (Yu-Taeger rat), 33991565 (Martella, dipraglurant rescue), 31320448 (Eskow Jaunarajs, cholinergic), 35699413 (Aïssa, cerebello-thalamic), 40902679 (Longueville 2025, cell-type-specific).

- **Biochemistry (Gα-olf):** PMIDs 2105931 (Jones, Golf is Gs-like stimulatory Gα), 15829631 (Von Dannecker, Ric-8B regulator).
- **Treatment evidence:**

P 20633181

(Novak, Class I/Level A botulinum toxin for cervical dystonia).
- **Computational:** Variant-class tabulation (`gnal_variant_spectrum.png`) — ~60% truncating vs ~40% missense, supporting haploinsufficiency.

Limitations & Future Directions

- No DYT25-specific prevalence/incidence, penetrance %, or QoL data; figures are extrapolated from broader adult-onset focal dystonia and small cohorts.
- Modifier genes and the basis of incomplete penetrance are unknown.
- GNAL-specific mechanistic data are largely rodent/in vitro; human evidence is from genetics plus network imaging of the (mostly non-genotyped) cervical-dystonia phenotype. There is no human neuropathology — the disease is functional/circuit-based, not degenerative.
- Mechanism-based therapies (mGlu5 NAM, cAMP/A2A modulators) remain preclinical and warrant clinical translation; natural-history and genotype–DBS-outcome studies are needed.
